



How Artificial Intelligence Impacts the US Labor Market

AI is often portrayed as a job killer, but new research co-authored by MIT Sloan associate professor Lawrence Schmidt paints a more nuanced picture.

- AI adoption leads to increased company growth in revenue, profits, employment, and profitability.
- Exposure to AI is greatest in higher-paying roles that involve information processing and analysis.
- In the 2014 – 2023 period, AI-exposed roles did not experience job losses relative to other roles, due to offsetting factors.

How AI is reshaping employment

The study used natural language processing to construct new measures of workers' task-level exposure to AI and machine learning from 2010 to 2023.

Key findings:

- When AI automates most of a job's responsibilities, the share of people in that role within a company falls by about 14%.
- When AI only automates *some* tasks, employment in that role can increase.
- Even workers in high-wage roles heavily exposed to AI — positions near the top of the pay scale — saw their share of total employment grow by about 3% over five years.



Firms that adopt AI don't necessarily need to shed workers; they can grow ... and use workers more efficiently than other firms.

Lawrence Schmidt
MIT Sloan Associate Professor



Which jobs shrank — and which grew

Exposure to AI is greatest in high-paying roles, which often involve tasks that AI can already do well, like information processing and analysis.

Roles	AI's Impact on Growth	Reason
Top-paying roles like management analysts, aerospace engineers, and computer scientists	Employment within firms fell by about 3.5% over five years.	These more lucrative roles are more likely to be in firms that adopt AI; the productivity gains that those firms realize more than offset the losses. As a result, the highest-paid jobs still make up a slightly bigger share of employment than those less exposed to AI.
Business, financial, architecture, and engineering jobs	These jobs shrank — by about 2% to 2.5% over five years.	These shrank — by about 2% to 2.5% over five years — because they have a high share of tasks that match what AI can do.
Legal jobs	Roles saw a 6.4% increase in employment.	These roles gained the most from AI. They face little direct impact from automation and are often in firms that use AI heavily.

Why even low-exposure jobs are at risk

But not all low-exposure roles are safe. **Even jobs with little direct impact from AI can shrink if their employers are slow to adopt the technology.**

- Food service jobs wilted relative to other jobs because employers that don't use AI grow more slowly, reducing demand for workers.
- Workers in occupations involving tasks amenable to AI but whose employers do not adopt AI are not necessarily better off, since the employers grow less rapidly than their AI-adopting peers.

Generative AI could change the equation

However, the study's data ends in 2023, just before the rapid rise of generative AI tools like ChatGPT. According to Schmidt, whether these patterns hold in the generative AI era is "the million-dollar question."

"Generative AI might reduce the gains from shifting work to people, but it could also make everyone more productive — a rising tide that lifts all boats," he said.

Either way, **AI's future impact will depend as much on management choices as on the technology itself.**

What employers can do

For Schmidt, the results highlight how much management decisions can shape AI's impact. The key to minimizing worker displacement from AI is to embrace the technology in the right way, he said.

To help companies get the most out of AI, Schmidt outlined a few priorities for business leaders:

- **Encourage hands-on use.** Do not wait until AI is rolled out across the company; give employees the chance to try different tools now. Early, active use builds skills and confidence.
- **Pick the right tools and use them well.** Make sure teams choose the versions and features that fit the work, and use them often enough to see real gains.
- **Think beyond efficiency.** Use AI not only to save time on routine work but also to open up new possibilities, such as tackling complex problems or generating fresh ideas.